

Developing ARM Targets Using GNU MCU Eclipse

If you are thinking of kickstarting the development of microcontroller targets, but don't have suitable hardware, or if you are worrying about hardware issues, here is the perfect solution for you. The GNU MCU Eclipse project comes with great features to overcome all these worries.



The GNU MCU Eclipse project, formerly known as GNU ARM Eclipse, offers a variety of plugins and components for the development of ARM targets for the Cortex-M series architecture. In this article, we'll discuss various components offered under this project, the supported features, and the installation and usage of each. We'll follow that up by building a simple project and emulating the Blinky code under the Qemu environment.

This project offers the following components for ARM development:

- MCU plugins to generate template code, as well as to build, flash and debug.
- Qemu, to emulate the generated outcome without the need of actual hardware, with good support for popular STM32 series targets.
- OpenOCD, to flash and debug code through JTAG probes and debugging hardware.
- Windows Build Tools, for the Linux-like make utility and basic commands like *rm* and *echo* used in *makefiles*.
- Packs manager, to install Keil/ARM CMSIS packs (in an experimental state, at present).

But the toolchain is not hosted under this project, and these plugins work well with standard toolchains designed

for supported architecture. Let's now take a tour of these components, their installation and usage.

Toolchain

You need a toolchain to build the code for the target architecture. For this, download the prebuilt binaries of GNU ARM Embedded Toolchain from developer.arm.com/open-source/gnu-toolchain/gnu-rm according to the host environment, i.e., Windows, Linux or Mac OS. This toolchain has a typical cycle of quarterly releases, and the latest release is 6-2017-q2-update which ships with GCC 6. From 6-2016-q4-major onwards, prebuilt binaries for 32-bit Linux have been dropped and only support for 64-bit Linux is available. However, one can build the toolchain from available sources for any host architecture. Releases up to v5-2016-q3 are also available at launchpad.net/gcc-arm-embedded, with binary support for 32-bit Linux only, and through PPA for Ubuntu distributions.

This toolchain provides various tools for development and the anatomy of generated files with the prefix 'arm-none-eabi-'. A few significant tools are:

- *arm-none-eabi-gcc*, which is a wrapper for various phases of C development